



AVINYA - The Hackathon Club

Problem Statements

Problem Statement 1:

Early Detection of Sepsis Using Real-Time Patient Data

Context:

Sepsis is a life-threatening condition caused by the body's extreme response to an infection. Early detection and timely intervention can significantly reduce mortality rates. However, sepsis diagnosis is often delayed due to the complexity of symptoms and variability in patient responses. Machine learning can help predict sepsis onset before clinical symptoms become critical, allowing for faster medical intervention.

Objective:

Develop a machine learning model to predict whether a patient will develop sepsis within the next 6 hours based on real-time physiological data from ICU (Intensive Care Unit) patients.

Dataset:

Participants will receive a dataset containing real-time patient monitoring data, including:

- Demographics: Age, gender, pre-existing conditions
- Vital Signs (Time-Series Data): Heart rate, respiratory rate, blood pressure, temperature, oxygen saturation
- Lab Test Results: White blood cell count, lactate levels, blood cultures, C-reactive protein
- Medication & Interventions: Antibiotic administration, fluid resuscitation, vasopressor usage
- Sepsis Label: Whether the patient developed sepsis within 6 hours

Challenges for Participants:

1. Time-Series Analysis: Process and analyze sequential data to identify patterns leading to sepsis onset.

2. **Data Imbalance:** Sepsis cases are rarer than non-sepsis cases, making classification difficult. Participants must implement techniques such as SMOTE (Synthetic Minority Over-sampling Technique) or class-weighted models.
3. **Feature Engineering:** Extract meaningful features from physiological data and combine static and time-dependent variables effectively.
4. **Real-Time Prediction:** Optimize models to provide early warnings with minimal computational delay.
5. **Explainability & Trustworthiness:** Develop interpretable models that provide clear justifications for their predictions to help doctors make informed decisions.

Evaluation Metrics:

- **Model Performance:** AUC-ROC, precision, recall, F1-score
- **Early Detection Accuracy:** The ability to predict sepsis before critical symptoms appear
- **False Alarm Rate:** Minimizing unnecessary alerts while maintaining sensitivity
- **Computational Efficiency:** Ensuring real-time applicability for ICU environments

Problem Statement 2:

AI-Powered Autonomous Financial Portfolio Optimization & Fraud-Resistant Algorithmic Trading

Context:

Financial markets are highly volatile, and traditional trading algorithms struggle to adapt to black swan events, adversarial attacks, and manipulative high-frequency trading (HFT) strategies. Portfolio optimization is a multi-objective problem, balancing returns, risk, liquidity, and ethical considerations while preventing fraud, market manipulation, and adversarial trading.

The challenge is to build an AI-driven trading and portfolio management system that dynamically adjusts investment strategies in real-time while resisting market manipulation, fraud, and adversarial shocks.

Objective:

Develop an AI-driven portfolio optimization and autonomous trading model that:

1. Maximizes risk-adjusted returns while adhering to investment constraints.
2. Detects and resists fraudulent trading patterns (pump-and-dump, spoofing, wash trading).
3. Adapts dynamically to extreme market conditions and news sentiment.
4. Maintains ethical constraints (ESG investing, regulatory compliance).

Dataset:

Participants will receive multi-source, real-time financial data, including:

1. Market & Trading Data:
 - Real-time stock, crypto, and forex prices (OHLCV).
 - Order book depth (bid-ask spreads, trade volumes).
 - Historical price data (for backtesting).
2. Alternative Data Sources:
 - Sentiment analysis on financial news and social media (Reddit, Twitter, Bloomberg).
 - ESG (Environmental, Social, and Governance) scores for ethical investing.
 - Central bank announcements, interest rate changes.
3. Fraudulent & Manipulative Trading Patterns (Labeled Data):

- Spoofing: Placing fake buy/sell orders to manipulate prices.
 - Pump-and-Dump: Artificially inflating asset prices.
 - Wash Trading: Self-trading to fake volume.
4. Macro & Risk Indicators:
- Volatility indexes (VIX).
 - Yield curve inversions, inflation trends.
 - Correlations between asset classes (equities, commodities, crypto).
5. Portfolio Constraints:
- Risk Management: Maximum drawdown, Sharpe Ratio, volatility limits.
 - Liquidity Constraints: Trade execution should not move market prices significantly.
 - Regulatory Compliance: Anti-money laundering (AML), KYC norms.

Challenges for Participants:

1. Multi-Objective Optimization:
- Maximize return while minimizing risk, transaction costs, and fraud exposure.
 - Optimize Sharpe ratio, Sortino ratio, and Maximum Drawdown (MDD).
2. Adversarial Robustness in Trading:
- Detect and resist algorithmic manipulation (spoofing, front-running).
 - Ensure resilience against flash crashes and market anomalies.
3. Reinforcement Learning-Based Autonomous Trading:
- Implement Deep Reinforcement Learning (DRL) agents for adaptive trading strategies.
 - Use real-time state-action-reward modeling with limited lookahead.
4. Explainability & Risk Transparency:
- Justify why trades are executed to meet regulatory requirements.
 - Interpret why assets are allocated dynamically in a portfolio.
5. Real-Time Decision Making:
- Implement high-speed trading strategies that execute without human intervention.
 - Ensure real-time fraud detection and prevention before trades execute.

Evaluation Metrics:

- Portfolio Performance:
 - Sharpe Ratio, Sortino Ratio: Risk-adjusted returns.
 - Calmar Ratio: Performance during drawdowns.
 - Max Drawdown (MDD): Maximum portfolio loss over time.
- Fraud Detection Accuracy:
 - Detecting spoofing, wash trading, pump-and-dump with low false positive rates.
- Adversarial Robustness:
 - Ability to maintain performance despite market shocks, manipulation, and volatility spikes.
- Computational Efficiency:
 - System should react in real-time with minimal execution delays.
- Explainability & Regulatory Compliance:
 - Justify trade decisions for auditability and legal transparency.

Expected Real-World Impact:

This AI-driven algorithmic trading system can:

- Protect financial markets from fraud, manipulation, and adversarial AI attacks.
- Optimize investments dynamically with reinforcement learning.
- Help hedge funds, banks, and retail traders make smarter financial decisions.

Problem Statement 3:

AI-Driven Climate Change Resilience, Disaster Prediction, and Autonomous Emergency Response System

Context:

Climate change is increasing the frequency and severity of extreme weather events such as hurricanes, wildfires, floods, and droughts. Traditional forecasting models struggle with non-linear climate interactions, microclimate effects, and the cascading impacts of extreme events. Additionally, most models do not integrate real-time emergency response optimization or policy-based intervention planning to reduce disaster impact.

To achieve full-scale resilience, an AI-powered system must predict extreme weather events, model climate-induced socio-economic risks, and autonomously optimize disaster response strategies in real-time using multi-modal data fusion.

Objective:

Develop an AI system that:

1. Predicts extreme weather events, environmental degradation, and multi-scale climate-related risks with high temporal and spatial resolution.
2. Simulates cascading climate impacts (e.g., wildfires causing air pollution, droughts leading to migration) to forecast secondary and tertiary disasters.
3. Generates real-time autonomous disaster response strategies, optimizing emergency resource allocation, evacuation planning, and infrastructure reinforcement.
4. Provides policy-driven recommendations to mitigate future climate disasters using reinforcement learning and climate scenario simulations.

Dataset:

1. Climate & Meteorological Data:
 - Satellite imagery (MODIS, Landsat, Sentinel, NOAA GOES, Himawari).
 - Historical climate records (temperature, precipitation, wind, sea level rise).
 - Real-time atmospheric and oceanic conditions (ENSO, Arctic ice changes, jet streams).
 - Aerosol and greenhouse gas concentration data from global monitoring stations.
2. Geospatial & Urban Resilience Data:

- High-resolution population density and urban infrastructure maps.
 - Building vulnerability indices for earthquakes, floods, and wildfires.
 - Traffic congestion, emergency exit routes, and real-time transportation data.
 - Real-time sensor networks (IoT-enabled environmental monitoring).
3. Socioeconomic & Human Impact Data:
- Climate migration patterns and refugee displacement data.
 - Agricultural crop yield and food supply chain disruptions.
 - Energy consumption, industrial emissions, and sustainability policy compliance.
4. Extreme Weather & Disaster Impact Data:
- Hurricane, flood, drought, wildfire, and heatwave occurrences.
 - Economic losses and insurance claim records for past disasters.
 - Social media and emergency call center logs to assess real-time distress signals.

Challenges for Participants:

1. Multi-Scale & Multi-Modal Forecasting:

- Combine short-term disaster prediction (e.g., hurricane landfall within 72 hours) with long-term climate scenario modeling (e.g., 50-year sea level rise impact).
- Integrate structured numerical data with unstructured satellite images, text reports, and IoT sensor feeds.
- Model real-time interactions between multiple climate systems (e.g., melting Arctic ice affecting monsoon rainfall).

2. Cascading Disaster Prediction & Risk Simulation:

- Identify interlinked climate hazards (e.g., drought → food shortages → migration → civil unrest).
- Use graph neural networks (GNNs) or knowledge graphs to model relationships between disasters.
- Predict indirect economic losses, such as supply chain failures and public health emergencies.

3. Adversarially Robust & Bias-Resistant AI Models:

- Address historical biases in climate models that under-represent developing nations.
- Ensure resilience against data poisoning attacks (e.g., fake sensor data injection).

- Develop uncertainty-aware models to handle missing climate data and sensor outages.

4. Autonomous Disaster Response & Resource Allocation Optimization:

- Implement reinforcement learning (RL) for emergency response planning, dynamically allocating medical supplies, shelter, and rescue teams.
- Optimize real-time evacuation routes based on real-world constraints (traffic, infrastructure collapse, human panic behavior).
- Develop AI-driven simulation environments to test policy interventions under different climate scenarios.

5. Computational Scalability & Real-Time Decision Making:

- Ensure AI models process petabytes of climate and geospatial data in real-time for fast decision-making.
- Implement distributed and edge AI for faster inference on remote disaster sites with limited connectivity.
- Use quantum-inspired algorithms to solve complex climate simulations and optimization problems.

Evaluation Metrics:

1. Extreme Weather Event Prediction Accuracy:

- Recall and precision on rare event classification (hurricanes, wildfires).
- Lead time improvement in disaster forecasting.

2. Disaster Impact Simulation Fidelity:

- Accuracy in predicting secondary climate disasters (e.g., floods after wildfires).
- Alignment of AI-generated scenarios with historical climate trends.

3. Resource Optimization Efficiency:

- Reduced emergency response time using AI-based deployment.
- Lower casualties and economic damage based on model recommendations.

4. Bias & Ethical AI Compliance:

- Fairness in disaster impact prediction across different socio-economic groups.
- Adversarial robustness and security testing against data tampering.

5. Computational Performance:

- Real-time decision-making capabilities with minimal latency.

- Scalability across global climate datasets with minimal hardware constraints.

Extreme Constraints to Increase Difficulty:

1. Limited Real-Time Data Access:

- AI must function with partial sensor failure or missing geospatial data.
- Model must handle dynamic uncertainty (e.g., unexpected atmospheric shifts).

2. Low-Resource and Developing Region Adaptability:

- System must work in regions with minimal climate monitoring infrastructure.
- Use synthetic data generation to improve model performance for underrepresented areas.

3. Dynamic Policy Adaptation:

- AI must adjust its recommendations based on evolving climate policies and legal frameworks.
- Model should simulate the economic impact of carbon taxes and sustainability laws.

4. Autonomous AI Deployment in Disaster Zones:

- Integrate AI models with drone-based remote sensing for disaster reconnaissance.
- Ensure AI functions with limited compute power and edge devices in crisis regions.

Expected Real-World Impact:

- Early warning systems for extreme weather events with improved accuracy and response time.
- AI-driven climate adaptation planning to reduce economic damage and loss of life.
- Smarter climate policy recommendations based on real-time disaster simulation.
- Scalable climate intelligence platform for governments, insurers, and humanitarian organizations.

***Note: One problem statement will be finalized, and it will be common for all participants. Everyone will work on the same problem.**

